

LLM Core Capabilities — Interview Notes (Concept-Clear)

Ye 8 core capabilities hain jo har LLM me evaluate ki jaati hain. Har benchmark in me se kisi ek (ya zyada) capability ko target karta hai — isliye benchmarks padhne se pehle in 8 capabilities ka clear hona zaroori hai.

1. Knowledge & Reasoning

Ye domain kya measure karta hai

Do cheezein:

- Model ne training ke dauraan **kitna factual knowledge** seekha hai.
- Kya wo **multiple pieces of information ko connect** karke, unke through reason karke sahi conclusion tak pahunch sakta hai.

Ismein kya aata hai

- **Factual recall across subjects** — biology, physics, chemistry, history, law, medicine, economics, philosophy jaise areas ke facts test karna. Ye model ne kya seekha hai uska sabse basic test hai.
- **Multi-step logical reasoning** — kya model kai facts ko sahi sequence me combine kar sakta hai. Sirf ek fact se answer nahi, balki reasoning ki poori chain follow karke final answer tak pahunchna.

Frontier labs isko alag se kyun evaluate karte hain

- Ye capability “model kitna intelligent hai” is question ka **proxy** maana jaata hai.
- Media aur general public me sabse zyada discuss hone wali capability bhi yahi hai — isliye labs ko dikhana padta hai ki unka model widely-recognized knowledge/reasoning benchmarks pe competitors ko match/beat kar raha hai.

Real-world relevance

- Research papers analyze karna, complex customer questions answer karna, technically accurate documents likhna, law/medicine/science/finance professionals ko assist karna.

2. Coding & Software Engineering

Ye domain kya measure karta hai

- Kya model **actually working code** likh sakta hai.
- Kya wo real software engineering environments (large codebases, multiple files, dev tools, repositories, complex workflows) ke andar kaam kar sakta hai.

Ismein kya aata hai

- **Function-level code generation** — natural-language instructions se individual functions likhna, edge cases handle karna, inputs validate karna, function interface follow karna.
- **Real-world bug fixing** — ek real GitHub issue aur existing repository diya jaata hai, model ko root cause identify karke correct patch banana hota hai, baki software ko tode bina.

- **Multi-file, long-horizon engineering tasks** — kai files me changes karna, system ke different parts ki dependencies samajhna, refactoring ya naye features add karna.
- **Command-line & system administration** — terminal ke through multi-step tasks (packages install karna, servers configure karna, diagnostic commands chalana, tools combine karna).
- **API & function calling** — sahi parameters ke saath API calls karna, errors handle karna, returned results ko sahi se process karna.

Frontier labs isko alag se kyun evaluate karte hain

- Better coding performance **developer productivity** directly improve karta hai aur bada business value create karta hai.
- Developers ek important customer group aur revenue ka major source hain, isliye labs coding benchmarks pe bahut compete karte hain.

Real-world relevance

- Code generation, debugging, code review, CI/CD automation, software maintenance, general software engineering work.
 - **Coding evaluations me leading models aur weaker models ke beech sabse bada performance difference dikhta hai.**
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3. Mathematics

Ye domain kya measure karta hai

- Kya model accurate **symbolic aur numerical reasoning** perform kar sakta hai.
- Maths evaluation ke liye especially useful hai kyunki final answer generally ya **correct hota hai ya incorrect** — sirf convincing sound karke response successful nahi ho sakta (jaise open-ended text me ho sakta hai).

Ismein kya aata hai

- **Grade-school maths & basic arithmetic reasoning** — multi-step word problems solve karna, natural language ko mathematical expressions me convert karna, sahi sequence me calculations karna.
- **Competition-level problem solving** — AIME jaise competitions ke problems, jisme creative reasoning, combinatorics, geometry, number theory, pattern recognition chahiye.
- **Undergraduate & graduate-level maths** — real analysis, abstract algebra, topology, probability theory, differential equations jaise advanced areas — long reasoning chains chahiye ho sakti hain.
- **Research-level mathematical reasoning** — mathematical research ke frontier ke qareeb ke problems, jahan expert mathematicians ko bhi solution tak pahunchne me kai din lag sakte hain.

Frontier labs isko alag se kyun evaluate karte hain

- Maths **genuine reasoning ka sabse cleanest test** deta hai.
- Ek plausible-sounding lekin galat answer ke liye yahan bahut kam gunjaish hai. Isse ye check karna aasan hota hai ki model **formal aur verifiable reasoning** kar raha hai ya sirf pattern-matching pe reliant hai.

Real-world relevance

- Scientific computing, financial modelling, engineering simulations, data analysis, koi bhi application jisko precise numerical results chahiye.
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4. Long Context

Ye domain kya measure karta hai

- Kya model **bahut lambe inputs** (kabhi-kabhi lakhon tokens) se information ko effectively use kar sakta hai.

Ismein kya aata hai

- **Single-fact retrieval from long contexts** — ek bahut lambe document ke andar kahin chhupi hui ek specific piece of information dhoondhna.
- **Multi-fact co-reference across long texts** — same person, entity, concept, ya constraint ko track karna jab wo document/conversation me kai jagah, ek doosre se door mention hui ho.
- **Aggregation over long content** — bade text me spread hui information ko summarize, count, compare, ya pattern-identify karna.
- **Code repository understanding** — large codebase navigate karna, files ke beech relationships samajhna, cross-file dependencies pe reason karna.

Frontier labs isko alag se kyun evaluate karte hain

- Context-window size ek important **marketing number** ban gaya hai: 128K → 200K → 1M tokens.
- Lekin bada advertised context window ye guarantee nahi karta ki model us poori information ko **effectively use** kar sakta hai.
- Labs ko dikhana padta hai ki unke claimed context capacity ko model genuinely use kar sakta hai.

Real-world relevance

- Legal contracts review karna, academic papers analyze karna, large codebases maintain karna, lambi customer-support conversations handle karna, documents ke bade collections ke saath kaam karna.
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5. Vision & Multimodal

Ye domain kya measure karta hai

- Kya model text ke saath **visual information ko samajh aur reason** kar sakta hai.

Ismein kya aata hai

- **Scientific chart & graph understanding** — axes, legends, data points, trends padhna. Model ko exact values extract karni pad sakti hain aur chart ka broader meaning bhi explain karna pad sakta hai.
- **Document & form understanding** — scanned documents, invoices, tables, forms process karna. Words ke saath-saath page pe information kaise arrange hui hai wo bhi samajhna zaroori hai.
- **GUI & screen understanding** — computer/mobile screens samajhna — buttons, menus, dialog boxes, input fields, aur application ka current state identify karna.

Frontier labs isko alag se kyun evaluate karte hain

- Vision se models un tasks ko handle kar paate hain jo text-only systems nahi kar sakte.
- Ye naye use-cases enable karta hai aur ek important area ban gaya hai jahan frontier models differentiate kar sakte hain — especially jab bahut saare text benchmarks saturate ho chuke hain.

Real-world relevance

- Invoices process karna, charts aur research figures analyze karna, software interfaces navigate karna, medical images padhna, engineering diagrams samajhna, visually complex documents ke saath kaam karna.
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6. Agentic & Tool Use

Ye domain kya measure karta hai

- Kya model sirf text generate karne ke bajaye **actions le sakta hai**.
- Ek agentic model internet browse kar sakta hai, APIs call kar sakta hai, files manipulate kar sakta hai, code execute kar sakta hai, software operate kar sakta hai, aur complex multi-step tasks complete kar sakta hai.
- Success ke liye model ko: apne actions **plan** karne padte hain, sahi tools **choose** karne padte hain, progress **track** karni padti hai, aur jab kuch galat ho jaaye to **recover** karna padta hai.
- Ye AI evaluation ke **next major frontiers** me se ek maana jaata hai.

Ismein kya aata hai

- **Web browsing & persistent research** — multiple websites navigate karna, links follow karna, information search karna, kai online sources se findings combine karna. Kuch tasks me dozens ya hundreds pages tak visit karni pad sakti hain.
- **Structured tool calling & API interaction** — external tools/APIs ko sahi se use karna — sahi function select karna, valid parameters pass karna, responses handle karna, calls ko chain karna, errors se recover karna.
- **Desktop & computer use** — real applications ko visually operate karna (clicking, typing) — word processors, spreadsheets, email clients, file managers ke across navigate karna.
- **Multi-turn task execution with planning** — bade goal ko chhote steps me todna, unhe sahi order me complete karna, intermediate results check karna, jab kuch fail ho to plan adjust karna.

Frontier labs isko alag se kyun evaluate karte hain

- Ek model ke paas bahut saara knowledge ho sakta hai lekin fir bhi wo us knowledge ko ek practical task complete karne ke liye use nahi kar paata.
- Agentic evaluation ye measure karta hai: **“Model kya jaanta hai”** vs **“Model actually kya kar sakta hai”** — is difference ko.
- Ye gap frontier models me bhi abhi bahut bada hai.

Real-world relevance

- Coding agents, research assistants, customer-service bots, data-entry automation, personal assistants — koi bhi workflow jahan AI ko sirf answer dene ke bajaye actions perform karne hain.

7. Safety & Alignment

Ye domain kya measure karta hai

- Kya model **responsibly behave** karne ke liye trust kiya ja sakta hai.
- Ye check karta hai: model harmful requests refuse karta hai ya nahi, deception/sycophancy avoid karta hai ya nahi, adversarial attacks resist karta hai ya nahi, aur user pressure daale tab bhi safety principles follow karta rehta hai ya nahi.

Ismein kya aata hai

- **Refusal of harmful content generation** — cybercrime, violence, harassment, misinformation, illegal activities involving requests ko model appropriately refuse karta hai ya nahi. Saath hi model itna over-cautious bhi na ho jaaye ki harmless/legitimate requests bhi refuse karne lage.
- **Resistance to adversarial attacks & jailbreaks** — jab users prompt injection, role-playing, encoded instructions, ya doosre adversarial methods se safety rules bypass karne ki koshish karein, tab bhi model apne safety rules maintain karta hai ya nahi.

- **Truthfulness vs sycophancy** — jab model user ke beliefs/expectations se disagree karta ho, tab bhi sahi answer deta hai ya nahi. Ek safe/reliable model sirf wahi nahi bolna chahiye jo user sunna chahta hai.
- **Cybersecurity capabilities** — cryptography, reverse engineering, digital forensics, exploitation jaise advanced cybersecurity tasks model perform kar sakta hai ya nahi. Ye potentially dangerous autonomous capabilities samajhne ke liye especially important hai.

Frontier labs isko alag se kyun evaluate karte hain

- Safety ek **regulatory requirement** bhi hai aur ek major **reputational concern** bhi.

Real-world relevance

- Koi bhi organization jo AI model public ko deploy karti hai, use ye test karna zaroori hai ki uske safety controls sahi se kaam kar rahe hain.
 - Ye domain directly affect karta hai: legal liability, regulatory compliance, brand reputation, user trust, responsible deployment.
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8. Instruction Following

Ye domain kya measure karta hai

- Kya model **exactly wahi karta hai jo user ne bola** — na zyada, na kam.
- Instructions me format, length, tone, style, structure, ya content se related requirements ho sakti hain.
- Ek model ka knowledge bahut achha ho sakta hai, lekin agar wo baar-baar simple instructions ignore karta hai to use karna frustrating ho jaata hai.

Ismein kya aata hai

- **Verifiable constraint satisfaction** — aise instructions follow karna jo automatically check ho sakein, jaise:
 - Word limit ke andar rehna.
 - Bullet points use karna.
 - Valid JSON produce karna.
 - Markdown structure follow karna.
 - Specific section headings use karna.
- **Handling ambiguous/incomplete instructions** — jab request underspecified ho to user ko probably kya chahiye ye samajhna, zaroorat pade to clarifying question poochna, aur jahan appropriate ho wahan reasonable defaults choose karna.

Frontier labs isko alag se kyun evaluate karte hain

- Instruction following strongly affect karta hai ki everyday use me model helpful lagta hai ya frustrating.
- Real production environments me **user satisfaction se sabse directly connected** capabilities me se ek yahi hai.

Real-world relevance

- Almost har practical LLM use instruction following pe depend karta hai. Examples:
 - Required format me reports generate karna.
 - Specific style rules ke hisaab se code likhna.
 - Structured JSON outputs produce karna.
 - Brand-tone requirements follow karna.
 - Coherent multi-turn conversations maintain karna.
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Quick Recap Table (Interview Cheat-Sheet)

#	Capability	Kya Test Karta Hai (One-Liner)	Real-World Use
1	Knowledge & Reasoning	Factual recall + multi-step logical reasoning	Research analysis, professional Q&A (law/medicine/finance)
2	Coding & Software Engineering	Working code + real repo/dev-environment tasks	Code gen, debugging, CI/CD, maintenance
3	Mathematics	Symbolic/numerical reasoning, verifiable answers	Scientific computing, financial modelling
4	Long Context	Effectively use very long inputs (100K+ tokens)	Legal contracts, large codebases, long conversations
5	Vision & Multimodal	Understand + reason over visual + text info	Invoices, charts, medical images, GUI screens
6	Agentic & Tool Use	Take actions (browse, call APIs, use tools), not just generate text	Coding agents, research assistants, automation
7	Safety & Alignment	Trustworthy, refuses harm, resists jailbreaks, truthful not sycophantic	Legal liability, compliance, brand trust
8	Instruction Following	Does exactly what's asked — format, length, tone, structure	Structured outputs, brand tone, multi-turn coherence

One-liner for interviews: *“LLM evaluation is organized around 8 core capabilities — Knowledge & Reasoning, Coding, Mathematics, Long Context, Vision & Multimodal, Agentic & Tool Use, Safety & Alignment, and Instruction Following — and every benchmark you’ll encounter is designed to target one or more of these, so knowing this map first makes learning any individual benchmark much easier.”*

LLM Benchmarks — Interview Notes (Concept-Clear)

1. Benchmark kya hota hai?

Benchmark = ek **standardized test** jo kisi LLM ki ek **specific capability** ko measure karta hai.

Model evals do type ke hote hain:

- **Standardized benchmarks** (fixed test, sab models pe same tareeke se apply hota hai)
- **Custom evals** (apni application ke hisaab se banaye jaate hain)

Har benchmark ek **research paper** se aata hai — jaise GSM8K, MMLU, SWE-Bench sab research papers hain jisme dataset, run config, scoring aur aggregation sab define hota hai.

2. Har Benchmark ke 4 Core Components

Example: **GSM8K** (Grade School Math, ~8K rows, 2021 me aaya)

(1) Dataset / Task

- Dataset = **Questions + Golden Answers** dono hote hain.
- GSM8K: grade-school level math word problems.
 - *Q: Natalia sold clips to 48 friends in April, half as many in May. Total?*
 - *A: 72*
- **Task** = model ko diya jaane wala instruction — “solve this math problem and give the answer.”

(2) Run Configuration

Ye batata hai ki evaluation ke waqt **settings kya rahengi** (dono models pe SAME hona chahiye, warna comparison invalid ho jaata hai).

(a) Prompt Construction

- **Zero-shot vs Few-shot** — zero-shot me sirf question diya jaata hai; few-shot me pehle kuch solved examples dikhaye jaate hain. GSM8K classically **8-shot** hota hai.
- **Chain-of-Thought (CoT) vs Direct** — CoT allow karne se model step-by-step soch ke answer deta hai, accuracy badh jaati hai. Direct me galti ka chance zyada.

(b) Decoding & Sampling Config

- **Temperature** — zero ke qareeb rakha jaata hai (deterministic output ke liye). Zyada temperature = creative/random answers.
- **Max tokens** — agar kam rakha to CoT beech me cut ho sakta hai (model answer complete nahi kar payega). Zyada bhi excess reasoning allow karta hai. Dono models ko same max tokens milna chahiye.

(c) Scoring Strategy & Tools

- **pass@1** — ek hi attempt, right ya wrong.
- **pass@k** — k attempts diye, agar ek bhi sahi aaya to correct maana jaata hai (lenient).
- **maj@k (majority@k)** — k baar same question puchha, jo answer sabse zyada baar aaya wahi final answer.
- **Tools allowed or not** — jaise Python interpreter ya web search allow karna score ko artificially boost kar sakta hai. GSM8K me tools NOT allowed. SWE-Bench jaise benchmarks me tools (GitHub access) allowed hote hain.

(3) Scoring Method

Do steps:

1. **Extraction** — model ka raw output (jo kisi bhi format me ho sakta hai — “The answer is 72” ya sirf “72”) se sahi answer nikalna. Usually **regex** ya structured output se hota hai.
2. **Comparison**:
 - **Closed-ended** (exact match, jaise $72 == 72$) — programmatic comparison, straightforward.
 - **Open-ended** (paragraph-type answers, no fixed answer) — **LLM-as-a-judge** ya human evaluation use hota hai.

(4) Aggregation

- Har question ka score (0 ya 1) hota hai; sabko combine karke final % nikala jaata hai.
- **Simple aggregation**: 920/1000 correct → 92% score.
- **Complex aggregation**: jaise MMLU (57 subjects) — har subject ka alag score hota hai, phir unhe combine karna hota hai — **simple average nahi**, kyunki subjects ke questions unequal hote hain, isliye **weighted mean** use hota hai.

3. Evaluation Process (Step-by-Step Loop)

Model evaluation basically ek **loop over the dataset** hai. Har item (question) ke liye 7 steps hote hain:

1. **Load the item** — dataset se ek question + uska golden answer uthao.
2. **Build the prompt** — few-shot examples inject karo, chat template apply karo, instructions add karo. Ab tumhare paas wo exact string (ya message list) ready hai jo model ko bheji jaayegi.
3. **Call the model** — pinned decoding config (temperature, max_tokens, etc.) ke saath. API model ke liye ye ek HTTP call hai; local model ke liye ek forward pass.
4. **Capture the raw output** — aur usse **verbatim log** karo. Ye important hai: agar score galat lage, to raw output hi batayega ki fail model ne kiya ya tumhare parser ne.
5. **Extract the answer** — regex ya parser use karke prose se sahi answer nikalo (jaise “The answer is 72” se sirf 72 nikalna).
6. **Grade** — extracted answer ko golden key se compare karo → result 1 (correct) ya 0 (incorrect).
7. **Store** — per-item result save karo, saath me wo prompt aur raw output bhi jisse ye result aaya (debugging/auditing ke liye zaroori).

Loop khatam hone ke baad — sab per-item scores ko **aggregate** karke final headline number (jaise “92% on GSM8K”) milta hai.

Pseudocode (isko samajhna interview ke liye kaafi hai):

```
results = []
for item in dataset:
    prompt = build_prompt(item, few_shot=8, template=chat_template)
    output = model.generate(prompt, temperature=0, max_tokens=512, stop=["\n\n"])
    pred = extract_answer(output)          # regex: last number
    score = (pred == item.gold)           # 1 or 0
    results.append({prompt, output, pred, item.gold, score})

accuracy = mean(r.score for r in results)
```

Problem: Ye loop dikhne me simple hai, lekin production-level implement karne me complex hai — error handling, retries, rate limits, batching, structured extraction, etc.

Eval Harness

Shuru me evaluation simple lagta hai: question lo, model ko bhejo, answer check karo, repeat karo. Lekin real me aur bhi cheezein handle karni padti hain:

- Final answer sahi se extract karna jab model **“The answer is C”**, **“C.”**, ya **“Option C”** likh de — sab ko same

treat karna.

- Benchmark ke **exact scoring method** se grade karna (har benchmark ka apna method hota hai).
- Hazaaron questions ko efficiently **batches** me bhejna.
- **Failed API requests ko retry** karna.
- **Rate limits** handle karna.
- **Eval harness** = ek ready-made system jo ye saari details already handle karta hai. Harness ko pehle se pata hota hai:
 - Benchmark (GSM8K, MMLU, etc.) ke questions kahan se milenge
 - Har question ko kaise format karna hai
 - Model ko kaise bhejna hai
 - Model ke answer ko kaise extract karna hai
 - Sahi hai ya galat kaise judge karna hai
 - Final benchmark score kaise calculate karna hai
- **Analogy**: Benchmark = exam paper, Eval Harness = **complete exam administration system** (instructions, answer-sheet format, checking rules, aur score calculation — sab kuch).
- Popular tools: **lm-evaluation-harness** (EleutherAI, industry standard), **Inspect**, **HELM**, **DeepEval** (zyada application-eval-focused, thoda zyada code likhna padta hai).
- lm-eval-harness se sirf ek **command** se pura benchmark run ho jaata hai — no manual loop coding needed.

4. Kaun Evaluation Karta Hai? (3 Stakeholders)

1. Frontier Labs (model builders)

Kaun: OpenAI, Anthropic, Google DeepMind, Meta, Mistral, DeepSeek, Qwen — jo bhi model ship kar raha hai.

Kyun evaluate karte hain:

- **Internally, during development** — ye invisible majority hai eval work ki. Evals guide karte hain pretraining decisions, post-training, aur ye decide karne me ki kaunsa checkpoint ship karna hai. Iske bina model development **blind** hai — jo measure nahi kar sakte wo improve nahi kar sakte.
- **Release gating** — decide karna ki model itna achha (aur safe) hai ki ship kiya jaaye.
- **Marketing** — launch post me jo benchmark table dikhti hai, wahi part hai jo public dekhta hai.

Numbers kaise read karein: Lab ke numbers ek **ceiling** hote hain — favorable conditions me measure kiye gaye, selection aur configuration me biased.

2. Third-Party Evaluators

Kaun: Independent organizations jinka product hi evaluation hai — **LMarena (Chatbot Arena)**, **Artificial Analysis**, **Epoch AI**, **Vals AI**, aur similar.

Kyun evaluate karte hain: Ye unka reason for existing hai. Ye models ko independently run karte hain aur comparisons publish karte hain. Inka business model hi neutral dikhne pe depend karta hai.

Numbers kaise read karein: Ye group AI engineer ke liye **sabse useful** hai — ek reason se: ye **har model ko khud, same conditions me** run karte hain. Ye wo bhi measure karte hain jo labs omit karte hain: **cost aur latency**.

3. Companies & AI Engineering Teams

- Labs **general capability** measure karte hain, third parties **cross-model comparability** measure karte hain.
- Lekin koi bhi ye measure nahi karta ki model tumhare **support emails ko sahi price pe sahi classify karta hai ya nahi** — apni specific use-case ke liye ye kaam khud AI engineers/companies ko karna padta hai.

Interview tip: Kabhi bhi sirf frontier lab ke published numbers pe blindly trust mat karo — pucho “evaluation kisne kiya, kis condition me?”

5. Benchmarks ki Limitations (IMPORTANT for Interviews)

(a) Benchmark Contamination

- Public benchmarks ka dataset internet pe available hota hai.
- Naye models jab pre-training me internet data scrape karte hain, to benchmark ke questions+answers bhi training data ka part ban sakte hain.
- Result: model "reasoning" nahi kar raha, balki **memorize** kar chuka hai answer.
- Solution:** private benchmarks ya dynamic benchmarks (jo regularly update hote hain) use karo.

(b) Benchmark Saturation

- Naya benchmark aata hai → shuru me sab models poorly perform karte hain (25-35%).
- Time ke saath models improve hote hain → scores 90%+ pe cluster ho jaate hain.
- Jab sab models ka score similar ho jaaye, to benchmark models ko **differentiate** nahi kar pata → benchmark **saturated** ho jaata hai aur retire kar diya jaata hai.
- Example: GSM8K, MMLU, SWE-Bench (old version) — sab saturated maane jaate hain.

(c) Configuration Gaming

- Labs apne model ko **favorable conditions** dete hain (jaise extra tools allow karna) aur competitor ko default conditions.
- Isse score me 5-10% ka artificial difference aa sakta hai.
- Isliye "config" ka pura transparency zaroori hai — agar nahi diya gaya, to number trust mat karo.

(d) Aggregation Gaming

- Multi-subject benchmarks (jaise MMLU) me sirf **average score** dikhaya jaata hai, individual subject-wise weaknesses chhupa di jaati hain.
- Example: Model Physics me achha, Economics me kharab — lekin sirf overall 85% dikhaya jaayega.
- Agar aapko specific domain application banani hai, to sirf aggregate score pe bharosa mat karo — subject-wise breakdown maango.

7. Knowledge Capability — 7 Famous Benchmarks ki Evolution Story

Knowledge capability ka matlab hai: LLM ne training ke dauraan kitna **world knowledge retain** kiya (uske weights/biases me kitna parametric knowledge chhupa hai). Ye sabse **fundamental capability** maani jaati hai — reasoning, coding jaise capabilities baad me "emergent behavior" ki tarah aayi.

Neeche di gayi **evolution story** yaad rakhna interview ke liye bahut useful hai — kyunki isse pata chalta hai ki har naya benchmark **kis problem ko solve karne ke liye** aaya.

Roadmap (Evolution Tree)

```
MMLU (2020) — "Mother of all benchmarks" — Breadth of Knowledge
|
├─> TruthfulQA (2021) → saturate → SimpleQA (2024) — Reliability/Truthfulness
├─> AGIEval (2023) — Human exams reuse karo
├─> GPQA (2023) — Depth of Knowledge (breadth ki jagah)
└─> MMLU-Pro (2024) — MMLU ki flaws fix karo
```

Sab saturate/near-saturate ho gaye → HLE (2025) — Breadth × Depth combined

(1) MMLU — Massive Multitask Language Understanding

“Mother of all benchmarks” (Sept 2020)

- **Kya test karta hai:** Sirf **Breadth of Knowledge** (depth nahi).
 - **Dataset:** 57 subjects, ~14,000 MCQs (4 options each). Questions GRE, USMLE, AP jaise real exams se aur textbooks se liye gaye.
 - **Task:** Question + 4 options do, model ko sahi option (A/B/C/D) batana hai.
 - **Scoring:** 2 tareeke — (a) model directly answer generate kare, ya (b) **log-probability** nikalo har option ki, jiski probability sabse zyada wahi answer. Dono me 2-3% ka difference aa sakta hai.
 - **Run config:** 5-shot, direct (no CoT), temperature 0, tools not allowed. **Prompt format bahut sensitive hai** — thoda sa wording change karne se score kaafi wave ho sakta hai (config gaming ka risk).
 - **History:** GPT-3 (2020) → 43.9% vs Human expert baseline 90%. GPT-4 (2023) → 86%. 2024 tak frontier models 86-92% range me cluster ho gaye → **saturated**.
 - **Kyun saturate hua:** Study me pata chala ~6.5% questions hi **galat/ambiguous** hain — isliye koi bhi 100% score nahi kar sakta, aur sab ek jagah cluster ho gaye.
 - **Kya measure NAHI karta:** Reasoning depth, calibration (model ko pata hai ki usse nahi aata?), open-ended generation, multilingual (English-only, Western curriculum-centric).
 - **Known issues:** Label errors (6.5%), high contamination (2020 se public hai), prompt format gaming.
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(2) TruthfulQA — Reliability Branch

(Sept 2021)

- **Problem jo solve karna tha:** MMLU breadth test karta hai, lekin ye guarantee nahi karta ki model **truthful** hai. Internet par misconceptions bhi hain (jaise “knuckle cracking se arthritis hota hai”) — bade models zyada data absorb karte hain to zyada misconceptions bhi absorb kar sakte hain.
 - **Famous finding:** Bade/zyada capable models is benchmark pe **zyada fail** karte the — “capability ≠ alignment” wala idea yahin se popular hua.
 - **Dataset:** 817 adversarial questions, 38 categories — common human misconceptions ke around.
 - **Scoring 3 tareeke:** Generation (model khud A/B/C/D print kare), **MC1** (sirf ek sahi answer maan ke log-prob compare), **MC2** (multiple sahi answers ho sakte hain, unki probabilities add karke score — ye default/standard method hai).
 - **Run config quirk:** “Zero-shot” bola jaata hai lekin actually 6 fixed unrelated example questions har baar prompt me bhejte hain (static hone ki wajah se zero-shot maana jaata hai).
 - **Kya measure nahi karta:** Factual recall (kyunki MCQ hai), “honesty under pressure” (kya model pressure daalne par jaan-boojh kar galat boleگا — ye MASK benchmark check karta hai).
 - **Saturation:** 2024 tak saturate ho gaya kyunki RLHF/instruction-tuning jaisi alignment techniques aane se models zyada truthful ho gaye. Interesting: iska **contamination training stage pe nahi, alignment stage pe hota hai** — ye dataset alignment fine-tuning data ka part ban jaata hai.
 - **Replaced by:** SimpleQA aur MASK (alag-alag directions me).
-

(3) AGIEval — Human Exams Reuse Branch

(April 2023)

- **Philosophy:** Naya benchmark banane ke bajaye, **existing human exams** (SAT, LSAT, Gaokao — China ka exam) hi le lo aur unhe LLM ko do. Isse do fayde: (a) naya dataset banane ki mehnat nahi, (b) **real human baseline** milta hai (estimated nahi, actually measured).
- **Dataset:** 20 exam sections, 8000+ questions. Bilingual — English + Chinese (pehla bilingual benchmark).
- **Format:** 18/20 exams MCQ, 2 exams short-answer.
- **Results:** GPT-4 launch pe 58%, Human average 67%, Human toppers 91%. Isse pata chala LLMs abhi bhi top humans se kaafi peeche hain.
- **Important interview point:** Jab news aati hai “LLM ne IIT-JEE me humans ko beat kar diya” — iska matlab ye NAHI hota ki model human intelligence ko surpass kar gaya. Ek exam crack karna ≠ multi-step reasoning, tool use, long-horizon tasks handle karna. Ye sirf **exam-specific knowledge** test karta hai.

- **Saturation:** 2024-25 tak frontier models human topper baseline (91%) ke qareeb pahunch gaye → saturated, ab use nahi hota.

(4) GPQA — Google-Proof Q&A (Depth Branch)

(Nov 2023)

- **Philosophy:** MMLU breadth test karta hai lekin questions **easy** the. Rather than breadth, chalo **depth** test karte hain — sirf 3 subjects (Physics, Chemistry, Biology) lekin **PhD-level, research-grade** questions.
- **“Google-proof” naam ka matlab:** Questions itne difficult hain ki agar ek non-expert ko Google access bhi diya jaaye aur 30 minutes diye jaayein, tab bhi wo solve nahi kar payega.
- **Quality control:** Har question **2 domain experts** ne validate kiya hai.
- **3 subsets:** Main (448), Extended (546), **Diamond (198)** — sabse difficult, aur jab log “GPQA score X%” bolte hain to generally Diamond subset ka hi hota hai.
- **History:** GPT-4 (2023) → 39%. GPT-4o (2024) → 56% (Diamond). OpenAI’s o1 (reasoning model) → 78%. OpenAI PhD experts hire karke test kiya → 69.7% (paper me claim tha 81.3% — discrepancy hai, isliye “PhD ko beat kar diya” wale marketing claims pe pura trust mat karo). Grok 4 (2025) → ~87%.
- **Kya measure nahi karta:** General graduate knowledge (sirf 3 science subjects), open-ended problem solving, **reasoning trace correctness** (agar model sahi answer tuk-ke se bhi pahunch jaaye to bhi marks milte hain — reasoning process valid hai ya nahi wo check nahi hota).
- **Known issues:** Bahut kam questions (198 diamond) → statistical confidence kam. PhD baseline number disputed. Contamination badh raha hai (2-3 saal purana). Sirf 3 domains cover karta hai.
- **Status (2026):** Near saturation — reasoning models 80%+ score karne lage hain.

(5) MMLU-Pro — “MMLU Rebuilt to Fix Its Flaws”

24.

- **Philosophy:** MMLU ek achha product tha, sirf uski flaws fix karo — naya banane ke bajaye upgrade karo.
- **3 major changes MMLU se:**
 1. **4 options → 10 options** — elimination-based guessing mushkil ho jaata hai, difficulty badh jaati hai.
 2. **Trivia/noisy questions hataye, reasoning-based questions add kiye** — ab sirf fact-recall se kaam nahi chalega.
 3. **57 subjects → 14 broad categories** — better balanced representation, kisi subject ka under/over-representation nahi.
- **Proof ye changes kaam kar rahe the:** Reasoning-capable models non-reasoning models se **20 points aage** the — matlab ab sochne ki zaroorat pad rahi hai.
- **Fun fact:** “MMLU-Redux” ek benchmark **nahi** hai — ye ek paper hai jisne prove kiya ki MMLU me 6-8% questions incorrect hain — isi finding ne MMLU-Pro banane ko trigger kiya.
- **Dataset:** 12,000 questions, 14 disciplines.
- **Kya measure nahi karta:** Open-ended generation, reasoning trace correctness, calibration. Reasoning models ko **unfair advantage** milta hai (explicitly reasoning-heavy questions design kiye gaye).
- **Status (2026):** Nearing saturation — models 80-90% range me.

(6) SimpleQA — Reliability Branch v2 (Replaces TruthfulQA)

(2024, by OpenAI)

- **Kyun special hai:** Ye **abhi bhi active** hai (saturate nahi hua) — poore session me discuss kiye gaye 7 benchmarks me se sabse “alive” wala.
- **Format innovation:** Ye MCQ NAHI hai — **short-answer, fact-seeking questions**. Options nahi diye jaate, khud answer likhna/generate karna hota hai. Isse difficulty automatically badh jaati hai (subjective exams MCQ se harder hote hain).
- **Dataset:** 4326 short factual questions — aise questions jin par us waqt ka state-of-the-art model (GPT-4) **fail** hua

tha.

- **Unique innovation — Calibration bhi measure karta hai:** 3 categories me grade karta hai — **Correct, Incorrect, Not Attempted** (“I don’t know”). Agar model “pass/I don’t know” bolta hai to ye batata hai ki model apne knowledge ke limits ke baare me aware hai (hallucinate nahi kar raha).
- **3 metrics:** (a) **Correct** — overall kitne sahi, (b) **Correct given attempted** — sirf attempt kiye gaye questions me accuracy, (c) **F-score** — dono ka harmonic mean.
- **Result:** Same model jo MMLU pe 88% score karta tha, SimpleQA pe sirf ~40% score kar paaya — kaafi bada difference.
- **Kya measure nahi karta:** Long-form factuality (sirf short answers), everyday factual reliability (dataset hi rare/hard questions se bana hai — GPT-4 ki failures se).
- **Known issues:** LLM-as-judge use hota hai grading ke liye (judge khud improve hota rehta hai, isliye 2-saal-purane results se compare nahi kar sakte). **Answer staleness** (jaise “current world #1 rugby player” — answer time ke saath change ho jaata hai). GPT-4 ke against **adversarial/selection bias** (dataset hi GPT-4 ki failures pe based hai — dusre models ke liye fair nahi ho sakta).

(7) HLE — Humanities’ Last Exam (Breadth × Depth Combined)

(Jan 2025, still ACTIVE)

- **Core idea:** MMLU se **breadth** liya, GPQA se **depth** liya — dono ko combine karke ek massive benchmark banaya.
- **Naam ka matlab:** “Thesis, not marketing” — idea ye hai ki agar models is benchmark ko crack kar lein (closed-ended, broad, expert-level, unambiguous), to closed-ended Q&A test karne ko aur kuch bacha nahi — evaluation ko phir **open-ended, agentic tasks** ki taraf move karna padega.
- **Scale:** 2500 questions, **100+ subjects** (classics se rocket engineering tak), 1000+ experts, 500+ institutions, 50 countries ne contribute kiya — massive collaborative effort (typically benchmark ek chhoti research team se aata hai, ye alag hai).
- **Format:** 80% short-answer (generate karna hai), 20% MCQ. **10% questions multimodal hain** (image-based) — pehli baar kisi benchmark me visual data bhi hai.
- **Anti-contamination measure:** Ek **public set** (2500 questions, released) aur ek **private set** (institute ke paas, kabhi public nahi hoga) — naye models ko private set pe bhi test kiya ja sakta hai, taaki contamination check ho sake.
- **Calibration bhi measure karta hai:** Model se answer ke saath confidence % bhi maangte hain (e.g. “80% confident”), phir RMS calibration error nikalte hain.
- **History:** Jan 2025 launch, initially models single-digit % pe the. Grok 4 → 24%, GPT-5 → 25%, Gemini → 38% (2025). Gemini 3 Pro → 38%+ (2026 tak).
- **Kya measure nahi karta:** Open-ended/agentic problem solving, everyday usefulness, vision (sirf 10% hai) aur multilingual (English-only) capabilities poori tarah nahi.
- **Known issues:** Kuch questions disputed the → original 3000 se 2500 reh gaye. LLM-as-judge grading errors. **Failure-filter selection bias** — dataset hi 2024 ke frontier models ki failures se bana hai, isliye general/daily knowledge ko represent nahi karta.
- **Status:** Currently **the** state-of-the-art benchmark for testing knowledge + reasoning + math combined.

Quick Recap Table (Interview Cheat-Sheet)

Benchmark	Year	Tests	Format	Status (2026)
MMLU	2020	Breadth of knowledge	4-option MCQ, 57 subj	Saturated
TruthfulQA	2021	Reliability/Truthfulness	MC1/MC2/Gen	Saturated
AGIEval	2023	Human-exam comparison	MCQ + short-answer	Saturated
GPQA	2023	Depth (Physics/Chem/Bio)	4-option MCQ	Near saturation
MMLU-Pro	2024	Breadth + light reasoning	10-option MCQ, 14 disc	Near saturation
SimpleQA	2024	Factuality + Calibration	Short-answer (no options)	Active
HLE	2025	Breadth × Depth + Calibration	80% short-answer, 20% MCQ, 10% multimodal	Active

One-liner for interviews: “Knowledge benchmarks evolved from MMLU (breadth) → branched into reliability

(TruthfulQA→SimpleQA), human-exam comparison (AGIEval), depth (GPQA), and MMLU repair (MMLU-Pro) → all converged into HLE, which combines breadth and depth and also tests calibration, explicitly designed so that once models saturate it, evaluation must move to open-ended agentic tasks.”

“A benchmark is a standardized test with 4 parts — dataset, run config, scoring method, and aggregation — used to measure a specific LLM capability. But benchmarks aren’t perfect: contamination, saturation, config gaming, and aggregation gaming mean you should never blindly trust published numbers — always check who evaluated, under what conditions, and whether subject-level breakdowns exist.”

LLM Leaderboards — Interview Notes (Concept-Clear)

1. LLM Leaderboards Kya Hain?

Definition: Ek LLM leaderboard ek **public ranking/comparison table** hai jo dikhata hai ki different LLMs ek **common set of evaluations** pe kaisa perform karte hain. Simply bolen to — benchmark ek exam hai, aur uska result kahin publish/display karna hi **leaderboard** hai.

Leaderboards Kyun Exist Karte Hain? (4 Reasons)

1. **Common reference se models compare karna** — sabko same exam diya, kaun top, kaun last, turant pata chal jaata hai. Isse **trust** bhi milta hai — kyunki agar OpenAI khud bole “hum sabse achhe hain” to utna trust nahi hoga (self-interest hai), lekin agar ek **neutral third-party** yehi result de to zyada credible lagta hai.
2. **Model selection guide karna** jab khud evals run karne ke resources na ho — 100s models khud test karna bahut costly hai (time + money), isliye leaderboard se seedha top candidates uthaye ja sakte hain.
3. **Saturation detect karna** — agar top models ek jagah cluster (jaise 92-94%) karne lage, to samajh jaate hain ki benchmark saturate ho raha hai.
4. **Naye models discover karna** — top 3-4 to hamesha same rehte hain (OpenAI, Google, Anthropic), lekin thoda neech scroll karne se naye/cheaper models mil jaate hain jo apne use-case ke liye kaam aa sakte hain.

Kaun Use Karta Hai Leaderboards? (5 Stakeholders)

1. **AI Engineers** — 100 candidates ko 5 tak shortlist karne ke liye. Task-specific leaderboard use karte hain (jaise math app banana hai to math leaderboard dekhenge).
2. **Frontier Labs** — competitive positioning aur **release gating** ke liye. Agar naya model competitor ke pichhle model ko bhi beat nahi kar pa raha, to release nahi karte — internal go/no-go signal hai. (*Interesting example: Google ka “Nano Banana” model stealth mode me leaderboard pe aaya tha bina naam reveal kiye — jab benchmarks pe achha perform kiya tabhi officially launch kiya gaya.*)
3. **Researchers** — ye check karne ke liye ki ek nayi technique genuinely help kar rahi hai ya nahi, ek shared baseline ke against.
4. **Policymakers & Safety Institutes** — capability trends track karte hain forecasting/regulation ke liye.
5. **Open-weight ecosystem** — **discovery** ke liye sabse underrated function hai. Ek 3-log ki chhoti lab bhi ek frontier model ke saath same table pe dikh sakti hai agar unka model kisi benchmark pe achha kare.

4 Types of LLM Leaderboards

(1) Benchmark-Specific Leaderboards

- Sirf **ek single benchmark** ke results pe ranking banate hain (jaise sirf MMLU, ya sirf HumanEval, ya sirf GSM8K, ya sirf GPQA).
- **Answer karta hai:** “Is benchmark pe kaunsa model sabse best hai?”
- **Limitation:** Bahut **narrow view** deta hai — overall model kaisa hai wo pata nahi chalta. Zyadatar famous benchmarks ka apna dedicated leaderboard hota hai (example: HLE ka apna leaderboard).

(2) Multi-Benchmark Leaderboards — *Sabse useful category*

- Multiple benchmarks aur evaluation dimensions ko combine karke ek **cumulative/overall score** dete hain — Knowledge, Reasoning, Math, Coding, Instruction-following, Data analysis, sab ek saath.
- Practical deployment factors bhi dikhaate hain: **cost per token, latency, output speed, context-window size, price-to-performance ratio**.
- **Answer karta hai:** “Kaunsa model capability + cost + performance ka strongest overall combination deta hai?”

- Examples: **LiveBench** (23 objective tasks across 7 categories, contamination-free), **Artificial Analysis** (dedicated leaderboards for intelligence, speed, cost — bahut exhaustive list, coding agents, speech, image, audio, hardware sab cover karte hain).

(3) Human-Preference-Based Leaderboards

- Yaha benchmark score nahi, **human votes** se ranking banti hai. User ko 2 anonymous models (A vs B) ke response dikhaye jaate hain, user decide karta hai kaunsa better hai. Bahut saare votes collect karne ke baad ranking banti hai.
- **Answer karta hai:** “Users kaunse model ke responses zyada prefer karte hain?”
- Measure karta hai: helpfulness, clarity, writing quality, creativity, tone, conversational ability.
- Example: **LMarena (Chatbot Arena)** — “battle mode” me anonymous models compare karte hain.
- **Limitation:** Human bias hota hai — log lambe, zyada confident, better-formatted, “agreeable” answers ko prefer karte hain, jo zaroori nahi ki **sabse accurate/safe/best-reasoned** ho. Voting population bhi shayad tumhare actual users ko represent na kare.

(4) Application-Specific Leaderboards

- Ek particular real-world domain/task ke around banaye jaate hain — jaise software bugs fix karna, API/tool calling, SQL query generation, medical questions, agentic tasks, long documents.
- **Answer karta hai:** “Is specific application ke liye kaunsa model/system best hai?”
- Example: **Berkeley Function-Calling Leaderboard (BFCL)** — sirf tool-calling capability ko rank karta hai. Multiple benchmarks use karta hai, lekin single domain me.

Leaderboards Ko Blindly Trust Kyun Nahi Kar Sakte (7 Reasons)

1. **Benchmark performance real applications me transfer nahi hoti** — benchmark questions clean, well-defined, self-contained hote hain. Real world me ambiguous requests, missing info, company-specific data, multi-turn conversations, tool failures, edge cases hote hain. (Analogy: Kaggle competitions jeetna $\neq$ achha real-world data scientist hona — Kaggle data bahut clean hota hai.)
2. **Benchmark contamination score inflate kar sakti hai** — public questions training data ka part ban sakte hain, isliye high score genuine capability, memorization, ya similar-questions-se-familiarity — teeno me se kuch bhi ho sakta hai.
3. **Models leaderboard ke liye over-optimize ho sakte hain (leaderboard overfitting)** — companies apne model ko specifically leaderboard pe achha dikhane ke liye tune karti hain (jaise LMarena pe jeetne ke liye “human-pleasing” responses generate karna sikhaya jaata hai — lambe, confident, chapalusi-wale answers). Isse related hai **Goodhart’s Law**: *“Jab ek measure target ban jaata hai, wo measure ke roop me kam useful ho jaata hai.”* (Analogy: agar ek car company sirf mileage improve karne pe focus kare, to overall car quality gir jaayegi.)
4. **Overall ranking subjective design choices pe depend karti hai** — multi-benchmark leaderboards ko decide karna padta hai: kaunse benchmarks include/exclude karne hain, scores kaise normalize karne hain, har capability ko kitna weight dena hai. Ye choices change karne se final ranking bhi change ho sakti hai — aur ye choices leaderboard ke ander **transparent nahi hoti** hamesha.
5. **Chhote score differences meaningful nahi hote** — Model A: 84.3, Model B: 84.1 — sirf 0.2 ka difference third rank aur fifth rank bana sakta hai. Ye difference prompt variation, small test set, ya evaluation noise se aa sakta hai. Agar **confidence intervals nahi diye gaye**, to adjacent ranks ko effectively **tied** samjho.
6. **Human-preference leaderboards me human bias hota hai** — jaisa upar discuss kiya, “better lagne wala” answer zaroori nahi “sabse sahi” answer ho.
7. **Scores stale, incomplete, ya self-reported ho sakte hain** — leaderboards me purane model versions ke results ho sakte hain, naye release-hue models missing ho sakte hain, discontinued models ke results reh sakte hain, ya company khud apne results submit kar sakti hai (unverified self-report).

AI Engineer Ki Tarah Leaderboards Kaise Read Karein (5-Step Framework)

Sabse important line: *“Leaderboards are a filtering tool, not a decision tool.”* — Leaderboard se seedha ek model select nahi karte, sirf top candidates **shortlist** karte hain.

Step 1 — Apni requirements pehle likho Leaderboard kholne se pehle clear karo: task type, latency budget, cost ceiling, context needs, deployment constraints (on-prem? open weights? data retention?). Ye karne se tum **rank #1 ki taraf blindly bias** nahi hoge.

Step 2 — Sahi board choose karo (famous wala nahi, apne task se match karne wala)

- Agents → SWE-bench Pro / Terminal-Bench / OSWorld
- Tool use → BFCL
- Chat → matching LMArena sub-arena
- RAG → MTEB (embedding models ki ranking)
- Budget-constrained → price aur throughput dikhaane wala board (jaise Artificial Analysis)

Step 3 — Board ko sahi se padho

- Exactly kya score ho raha hai? (Answer key, test suite, human vote, ya LLM judge?)
- Model kisne run kiya? (Independent evaluator ya vendor self-report?)
- Kis inference compute pe? (Reasoning-effort settings score aur bill dono change karti hain)
- Eval set kitna purana hai? (Static + public = contamination ka risk maano)
- Kya saturated hai? (Top 10 agar 2-3 points ke andar hain, to samajh lo ho chuka hai)
- Confidence intervals kahan hain? (Na ho to adjacent ranks ko tied maano)
- Agar composite hai — weights kya hain? (Unpublished weights = unpublished opinion)

Step 4 — 3-5 candidates tak cut karo Hard filters pehle apply karo: license, price ceiling, latency, apne region me availability.

Step 5 — Apna khud ka eval chalao (yahi asli decision hota hai) Apne real traffic se 50-few hundred examples lo — ye kisi bhi public benchmark se better hota hai. Quality, cost, latency, aur failure behaviour pe score karo.

One-liner for interviews: “Leaderboards are a filtering tool, not a decision tool — use them to narrow 100 models down to 3-5 candidates that match your task’s leaderboard type, then always run your own evaluation on your real traffic before making the final call.”
